

AIoT GREENHOUSE MANAGEMENT SYSTEM

Data Model: JSON Schema, Entity-Relationship & Data Dictionary

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Docente: Juan Antonio Castro Silva

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1. System Overview

The AIoT Greenhouse Management System is a full-stack platform to monitor, analyze, and automate greenhouse and hydroponic crop management. It integrates IoT sensor devices (or a software simulator) with a Machine Learning classification model to provide real-time environmental assessment and early-warning alerts.

The data model is composed of 8 entities organized into 4 functional modules:

- Module 1 - Identity & Access: users
- Module 2 - Physical Infrastructure: zones, devices
- Module 3 - Sensor & AI: sensor_readings, predictions
- Module 4 - Agronomy & Alerts: crop_types, crops, alerts

2. Entity-Relationship Summary

Each entity and its primary / foreign key relationships:

- users -> user_id (PK)
- zones -> zone_id (PK)
- devices -> device_id (PK) | FK: zone_id, registered_by
- sensor_readings -> reading_id (PK) | FK: device_id
- crop_types -> crop_type_id (PK)
- crops -> crop_id (PK) | FK: zone_id, crop_type_id, created_by
- alerts -> alert_id (PK) | FK: device_id, reading_id, assigned_to
- predictions -> prediction_id (PK) | FK: device_id, reading_id

2.1 Relationship Table

From Table	To Table	Cardinality	Description
users	devices	1 : N	Admin user registers many IoT devices
users	crops	1 : N	Operator creates many crop batches
users	alerts	1 : N	Alerts are optionally assigned to an operator
zones	devices	1 : N	A zone contains one or more sensor devices
zones	crops	1 : N	A zone hosts one or more crop batches
devices	sensor_readings	1 : N	Each device generates many time-series readings
devices	alerts	1 : N	A device can trigger many alerts over time
devices	predictions	1 : N	AI predictions are linked to the source device
sensor_readings	alerts	1 : 0..N	A reading may trigger zero or more alerts
sensor_readings	predictions	1 : 1	Each prediction is based on exactly one reading snapshot
crop_types	crops	1 : N	A crop species can have many active batches

3. Data Dictionary

Legend: PK = Primary Key | FK = Foreign Key | -- = Not applicable

3.1 Table: users

Stores system accounts. Three roles: admin (full access), operator (devices+crops), viewer (read-only).

Field	Data Type	PK/FK	Nullable	Default	Description / Constraints
user_id	SERIAL	PK	No	auto	Unique auto-incremented identifier
username	VARCHAR(80)	--	No	--	Unique login name. Lowercase alphanumeric.
email	VARCHAR(120)	--	No	--	Unique email. Used for login and notifications.
password_hash	VARCHAR(256)	--	No	--	bcrypt hash. Plain-text password is never stored.
full_name	VARCHAR(150)	--	No	--	User display name for the UI
role	VARCHAR(20)	--	No	viewer	Values: admin operator viewer
is_active	BOOLEAN	--	No	TRUE	FALSE disables login without deleting the record
created_at	TIMESTAMP	--	No	CURRENT_TIMESTAMP	Record creation timestamp (UTC)
updated_at	TIMESTAMP	--	No	CURRENT_TIMESTAMP	Last modification timestamp (UTC)

3.2 Table: zones

Represents physical sections of the greenhouse (e.g. Zone A - Leafy Greens). Groups devices and crops.

Field	Data Type	PK/FK	Nullable	Default	Description / Constraints
zone_id	SERIAL	PK	No	auto	Unique zone identifier
name	VARCHAR(100)	--	No	--	Unique human-readable name
description	TEXT	--	Yes	NULL	Optional description of the zone purpose
area_m2	DECIMAL(8,2)	--	Yes	NULL	Zone area in square meters. Must be > 0.
is_active	BOOLEAN	--	No	TRUE	FALSE when zone is temporarily out of service
created_at	TIMESTAMP	--	No	CURRENT_TIMESTAMP	Record creation timestamp (UTC)

3.3 Table: devices

Each IoT sensor node or simulated device. Belongs to one zone, registered by one admin.

Field	Data Type	PK/FK	Nullable	Default	Description / Constraints
device_id	SERIAL	PK	No	auto	Unique device identifier
zone_id	INTEGER	FK	No	--	References zones.zone_id
registered_by	INTEGER	FK	No	--	References users.user_id (admin who registered it)
name	VARCHAR(100)	--	No	--	Descriptive name (e.g. Sensor Node ZA-01)
serial_number	VARCHAR(60)	--	Yes	NULL	Physical ID. NULL for simulated devices.
device_type	VARCHAR(20)	--	No	simulated	Values: sensor_node actuator gateway simulated
status	VARCHAR(20)	--	No	online	Values: online offline error maintenance
firmware_version	VARCHAR(20)	--	Yes	NULL	Firmware version string (e.g. v2.3.1)
last_seen_at	TIMESTAMP	--	Yes	NULL	Last successful communication timestamp
created_at	TIMESTAMP	--	No	CURRENT_TIMESTAMP	Record creation timestamp (UTC)

3.4 Table: sensor_readings

Core time-series table. Every sensor measurement is one row. BIGSERIAL for high-volume. Indexed on device_id + recorded_at.

Field	Data Type	PK/FK	Nullable	Default	Description / Constraints
reading_id	BIGSERIAL	PK	No	auto	Large auto-incremented ID for high-volume inserts
device_id	INTEGER	FK	No	--	References devices.device_id
temperature	DECIMAL(5,2)	--	Yes	NULL	Air temperature in Celsius. Range: -10 to 60.
humidity	DECIMAL(5,2)	--	Yes	NULL	Relative humidity in %. Range: 0 to 100.
ph	DECIMAL(4,2)	--	Yes	NULL	Nutrient solution pH. Range: 0 to 14.
light_lux	DECIMAL(10,2)	--	Yes	NULL	Light intensity in lux. Range: 0 to 150,000.
co2_ppm	DECIMAL(8,2)	--	Yes	NULL	CO2 concentration in parts per million.
soil_moisture	DECIMAL(5,2)	--	Yes	NULL	Substrate moisture in %. Range: 0 to 100.
recorded_at	TIMESTAMP	--	No	CURRENT_TIMESTAMP	Exact time the reading was captured (UTC)
is_simulated	BOOLEAN	--	No	FALSE	TRUE when generated by the software IoT simulator

3.5 Table: crop_types

Reference catalog of crop species with validated optimal growing parameters. Used to compute alert thresholds and AI labels.

Field	Data Type	PK/FK	Nullable	Default	Description / Constraints
crop_type_id	SERIAL	PK	No	auto	Unique crop species identifier
name	VARCHAR(100)	--	No	--	Common name (e.g. Lettuce). Must be unique.
scientific_name	VARCHAR(150)	--	Yes	NULL	Latin binomial name (e.g. Lactuca sativa)
description	TEXT	--	Yes	NULL	Optional growing notes
temp_min	DECIMAL(5,2)	--	No	--	Minimum viable temperature in Celsius
temp_max	DECIMAL(5,2)	--	No	--	Maximum viable temperature in Celsius
temp_optimal	DECIMAL(5,2)	--	No	--	Ideal temperature. Must be between min and max.

Field	Data Type	PK/FK	Nullable	Default	Description / Constraints
humidity_min	DECIMAL(5,2)	--	No	--	Minimum relative humidity in %
humidity_max	DECIMAL(5,2)	--	No	--	Maximum relative humidity in %
ph_min	DECIMAL(4,2)	--	No	--	Minimum acceptable pH of nutrient solution
ph_max	DECIMAL(4,2)	--	No	--	Maximum acceptable pH of nutrient solution
light_min_lux	DECIMAL(10,2)	--	Yes	NULL	Minimum light intensity in lux
light_max_lux	DECIMAL(10,2)	--	Yes	NULL	Maximum light intensity before stress in lux
growth_days	INTEGER	--	Yes	NULL	Approximate days from sowing to harvest
created_at	TIMESTAMP	--	No	CURRENT_TIMESTAMP	Record creation timestamp (UTC)

3.6 Table: crops

Active and historical crop batches. Links zone + species + operator, enabling environmental-to-yield correlation.

Field	Data Type	PK/FK	Nullable	Default	Description / Constraints
crop_id	SERIAL	PK	No	auto	Unique batch identifier
zone_id	INTEGER	FK	No	--	References zones.zone_id
crop_type_id	INTEGER	FK	No	--	References crop_types.crop_type_id
created_by	INTEGER	FK	No	--	References users.user_id (operator who created it)
batch_code	VARCHAR(50)	--	Yes	NULL	Traceability code (e.g. LETT-ZA-2026-01)
quantity	INTEGER	--	No	--	Number of plants. Must be > 0.
planted_at	DATE	--	No	--	Date seeds or seedlings were planted
expected_harvest_at	DATE	--	Yes	NULL	Target harvest date
actual_harvest_at	DATE	--	Yes	NULL	Actual harvest date. NULL until harvested.
status	VARCHAR(20)	--	No	germinating	Values: germinating growing flowering harvested failed
notes	TEXT	--	Yes	NULL	Agronomic notes or incident observations
created_at	TIMESTAMP	--	No	CURRENT_TIMESTAMP	Record creation timestamp (UTC)

3.7 Table: alerts

Auto-generated when sensor readings exceed crop thresholds, or when the AI model predicts warning or critical condition.

Field	Data Type	PK/FK	Nullable	Default	Description / Constraints
alert_id	SERIAL	PK	No	auto	Unique alert identifier
device_id	INTEGER	FK	No	--	References devices.device_id
reading_id	BIGINT	FK	Yes	NULL	References sensor_readings. NULL for prediction alerts.
assigned_to	INTEGER	FK	Yes	NULL	References users.user_id (operator assigned to resolve)
alert_type	VARCHAR(30)	--	No	--	Values: temperature humidity ph light co2 soil_moisture device_offline prediction
severity	VARCHAR(10)	--	No	medium	Values: low medium high critical
message	VARCHAR(500)	--	No	--	Human-readable description of the alert condition
measured_value	DECIMAL(10,2)	--	Yes	NULL	Actual sensor value that triggered the alert
threshold_value	DECIMAL(10,2)	--	Yes	NULL	Limit that was exceeded
status	VARCHAR(20)	--	No	open	Values: open acknowledged resolved dismissed
resolved_at	TIMESTAMP	--	Yes	NULL	Timestamp when resolved. NULL while still open.
created_at	TIMESTAMP	--	No	CURRENT_TIMESTAMP	Alert generation timestamp (UTC)

3.8 Table: predictions

Stores each AI inference for full traceability: input features, model version, confidence, and full probability vector.

Field	Data Type	PK/FK	Nullable	Default	Description / Constraints
prediction_id	SERIAL	PK	No	auto	Unique prediction identifier
device_id	INTEGER	FK	No	--	References devices.device_id
reading_id	BIGINT	FK	No	--	References sensor_readings.reading_id (input snapshot)
model_name	VARCHAR(100)	--	No	--	Model identifier (e.g. RandomForestClassifier_v1)
model_version	VARCHAR(20)	--	No	--	Semantic version of serialized model file (e.g. 1.0.0)

Field	Data Type	PK/FK	Nullable	Default	Description / Constraints
predicted_class	VARCHAR(20)	--	No	--	Values: optimal warning critical
confidence	DECIMAL(5,4)	--	No	--	Probability of predicted class. Range: 0.0000 to 1.0000.
raw_probabilities	JSONB	--	Yes	NULL	Full probability JSON: {optimal, warning, critical}
input_features	JSONB	--	Yes	NULL	Sensor values used as model input (audit trail)
created_at	TIMESTAMP	--	No	CURRENT_TIMESTAMP	Inference timestamp (UTC)

4. API Endpoints Catalog

REST API provided by the Flask backend. All responses are JSON. Authentication via JWT Bearer token (except /auth/login).

Auth	Endpoint	Description
POST	/auth/login	Login and receive JWT token
POST	/auth/logout	Invalidate current token
GET	/auth/me	Return current authenticated user

Users	Endpoint	Description
GET	/users	List all users (admin only)
POST	/users	Create new user (admin only)
GET	/users/{id}	Get user by ID
PUT	/users/{id}	Update user data
DELETE	/users/{id}	Deactivate user (soft delete)

Zones	Endpoint	Description
GET	/zones	List all zones
POST	/zones	Create zone
GET	/zones/{id}	Get zone detail
PUT	/zones/{id}	Update zone

Devices	Endpoint	Description
GET	/devices	List all devices
POST	/devices	Register new device
GET	/devices/{id}	Get device detail
PUT	/devices/{id}/status	Update device status

Readings	Endpoint	Description
GET	/readings	List readings (paginated)
POST	/readings	Insert new reading
GET	/readings/latest	Latest reading per device
GET	/readings/device/{id}	Readings for a device

Crop Type s	Endpoint	Description
GET	/crop-types	List all crop types
POST	/crop-types	Create crop type (admin)
GET	/crop-types/{id}	Get crop type detail

Crop s	Endpoint	Description
GET	/crops	List all active crops
POST	/crops	Create crop batch
GET	/crops/{id}	Get crop detail
PUT	/crops/{id}	Update status or notes
GET	/crops/zone/{id}	Crops in a zone

Alert s	Endpoint	Description
GET	/alerts	List all alerts (filterable)
GET	/alerts/open	Open alerts only
PUT	/alerts/{id}/acknowledge	Mark acknowledged
PUT	/alerts/{id}/resolve	Mark resolved

AI Predict	Endpoint	Description
POST	/predict	Run AI inference on sensor values
GET	/predictions	List all predictions
GET	/predictions/device/{id}	Predictions for a device

IoT Simulator	Endpoint	Description
POST	/simulator/start	Start the IoT data simulator
POST	/simulator/stop	Stop the simulator
GET	/simulator/status	Get simulator state

5. Seed Data Summary

The following reference data is inserted on initial database setup:

Users (3): admin, operator1, viewer1

Zones (4): Zone A - Leafy Greens | Zone B - Tomatoes | Zone C - Herbs | Zone D - Seedlings

Devices (5): One simulated sensor node per zone (ZA-01, ZA-02, ZB-01, ZC-01, ZD-01)

Crop Types (7): Lettuce, Tomato, Basil, Spinach, Strawberry, Mint, Bell Pepper

Crops (4): Active batches at various growth stages — germinating, growing, flowering

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